Towards a Practical Geometry Correcting Trust Region Algorithm with Theory

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Outline

1. Review + Motivation
2. Algorithm
3. Results

Setting the Scene

- Analyzed complexity of trust region (TR) methods with geometry correcting (GC) and randomized subspaces
- Worst case oracle oracle complexity of GC same as basic TR: $\mathcal{O}\left(\frac{n^2}{\varepsilon^2}\right)$
- Improved oracle complexity when using randomized subspace: $\mathcal{O}\left(\frac{np}{\varepsilon^2}\right)$

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Goal:

- Improve the practicality and flexibility of the framework
- Provide a concrete implementation and numerical results
- Provide theoretical guarantees in terms of complexity

Overview of Changes

- Addition of a self-correcting geometry step
  - Idea: trial step not discarded and may improve geometry
- Maintain two sample sets:
  1. $\mathcal{Y}$ where good geometry is maintained
  2. $\mathcal{Z}$ which holds the remaining interpolation points
  - Idea: Points in $\mathcal{Y}$ are "kicked" to $\mathcal{Z}$. Allow the use of arbitrary Lagrange polynomial bases with quadratic interpolation.
- Merged criticality step with acceptance step (also in GC-TR)

Note: This work will focus on linear Lagrange polynomials

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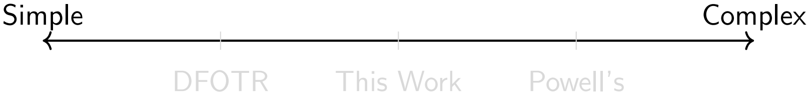
Big Picture



Sample Set Operations

1. Replacing far-away points
2. Poisedness correction (max Lagrange Poly. in ball)
3. Self-correction (max Lagrange Poly. at trial step)

- DFOTR: (1)
- This work: (1), (2), (3), with Linear Lagrange Poly.
- Powell's: (1), (2), (3) + engineering, with Quad. Lagrange Poly.

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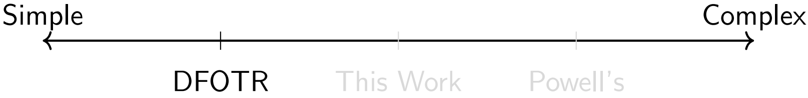

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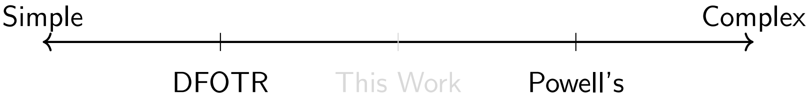

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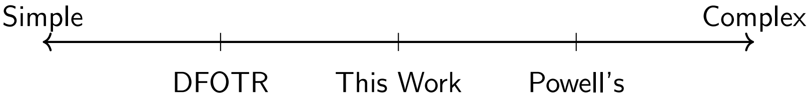

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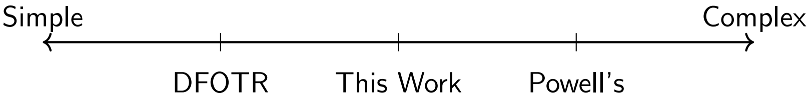

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Note: DFOTR and Powell's have no known complexity results.

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Initialization

Inputs:

- $f(x) = \phi(x)$, [Deterministic Zeroth-order oracle]
- $p = \frac{(n+1)n}{2}$, where n is the dimension

Initialize:

- Set $\mathcal{Y}_0$ such that $|\mathcal{Y}_0| = n$: [Geometry/Interpolation Set]
- Set $\mathcal{Z}_0$ such that $|\mathcal{Z}_0| \leq p$: [Interpolation Set]
- x_0: [Initial point]

Model Construction

Combine $\mathcal{Y}_k \cup \mathcal{Z}_k$ and fit a quadratic model as usual. That is

$$m_k(x_k + s) = \phi(x_k) + g_k(x_k)^\top s + \frac{1}{2} s^\top H_k(x_k) s$$

subject to the interpolation condition

$$m_k(s) = f(x_k + s) - f(x_k), \quad \forall s \in \mathcal{Y}_k \cup \mathcal{Z}_k$$

Note 1: In the numerics, for the under-determined case, we use minimum Frobenius norm model.

Model Construction - Bounded Model Hessians

Solve:

$$\min_{g,H} \sum_{z \in \mathcal{Z}_k} \left(f(x_k) + g^\top z + \frac{1}{2} z^T H z - f(x_k + z) \right)^2$$

$$\text{subject to: } \|H\|_F \le K$$

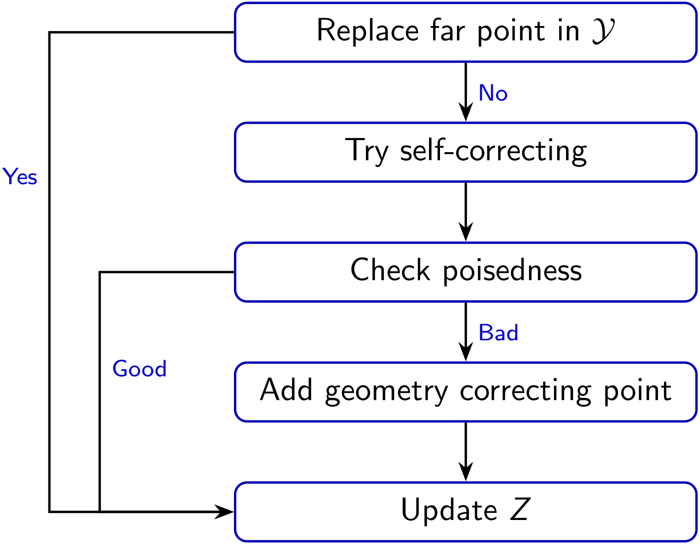
$$g^\top y + \frac{1}{2} y^\top H y = f(x_k + y) - f(x_k), \quad \forall y \in \mathcal{Y}_k$$

- Interpolate $\mathcal{Y}$ (strictly)
- Bounds Hessian matrix
- Tries to interpolate $\mathcal{Z}$ (regression)

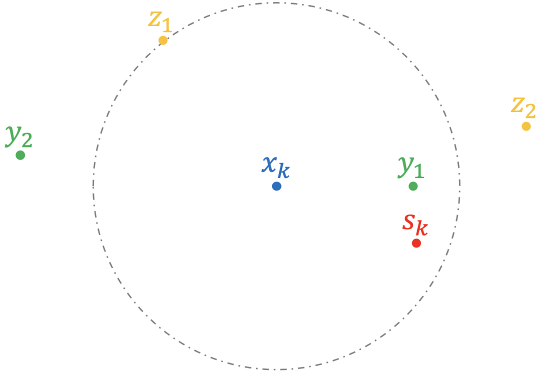
Pseudocode

for $k = 0, 1, ...$ **do**
    Build a quadratic model $m_k(x_k + s)$
    Compute trial step s_k and ratio ρ_k
    if *Successful Iteration:* $\rho_k \geq \eta_1$ *and* $\|g_k\| \geq \eta_2 \Delta_k$ **then**
        Run "successful" subroutine.
    else
        Run GC subroutine to improve geometry of $\mathcal{Y}$.
        if *any GC step is made* **then**
            Move appropriate points in $\mathcal{Y}$ to $\mathcal{Z}$.
        else
            Geometry is good, shrink Δ_k.
        end
    end
end

GC Subroutine Flow

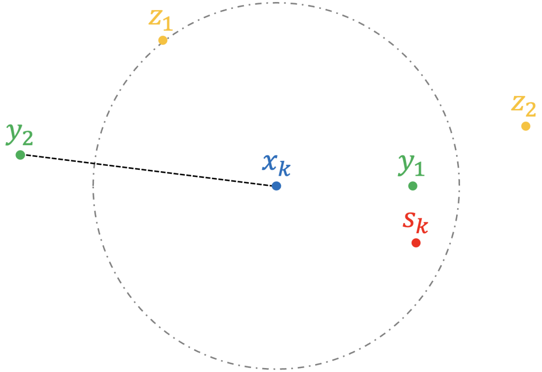


Unsuccessful GC Subroutine



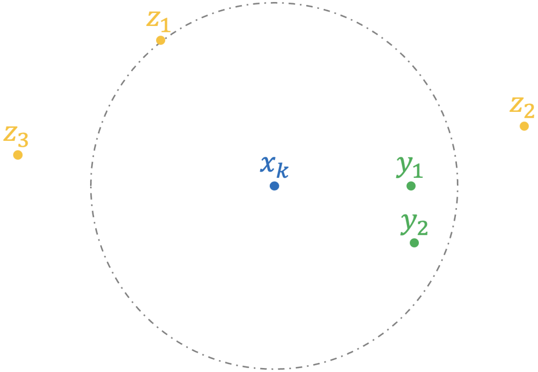
Unsuccessful step

Unsuccessful GC Subroutine



Replace far points in $\mathcal{Y}$

Unsuccessful GC Subroutine

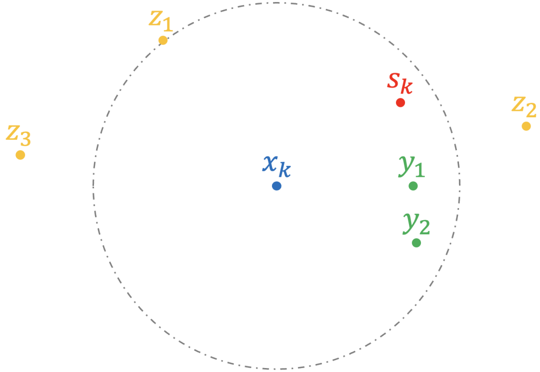


Update $\mathcal{Z}$

(Underdetermined)

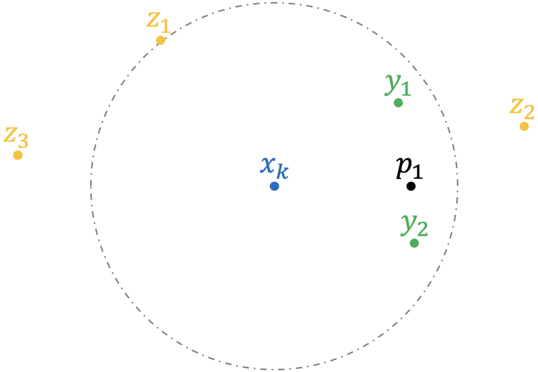
Ready to refit model

Unsuccessful GC Subroutine



Unsuccessful step.
Self-correcting:
$v = \max_{i=1,2} |\ell_i(s_k)|$

Unsuccessful GC Subroutine

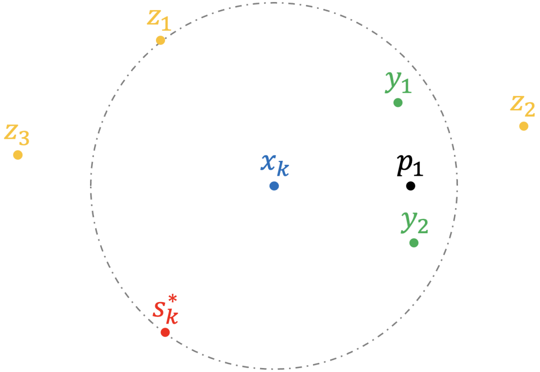


Self-correcting:
$v = \max_{i=1,2} |\ell_i(s_k)|$

If $v > \Lambda$ hold $p_1 = y_1$ and update $\mathcal{Y}$.

Otherwise keep $\mathcal{Y}$ and hold s_k.

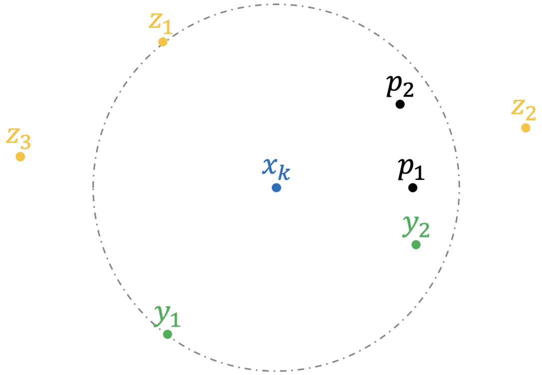
Unsuccessful GC Subroutine



Check poisedness:

$$v = \max_{i=1,2, s \in B(x_k, \Delta_k)} |\ell_i(s)|$$

Unsuccessful GC Subroutine



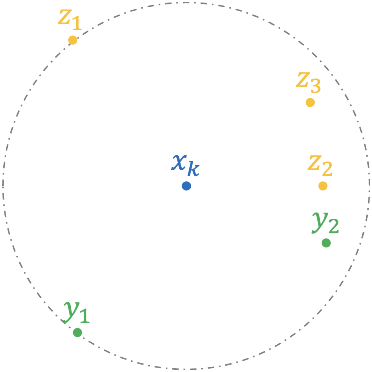
Check poisedness:

$$v = \max_{i=1,2, s \in B(x_k, \Delta_k)} |\ell_i(s)|$$

If $v > \Lambda$, Add GC point: hold $p_2 = y_2$, **evaluate** $f(s_k^*)$, update $\mathcal{Y}$

Otherwise geometry is good

Unsuccessful GC Subroutine



Update $\mathcal{Z}$

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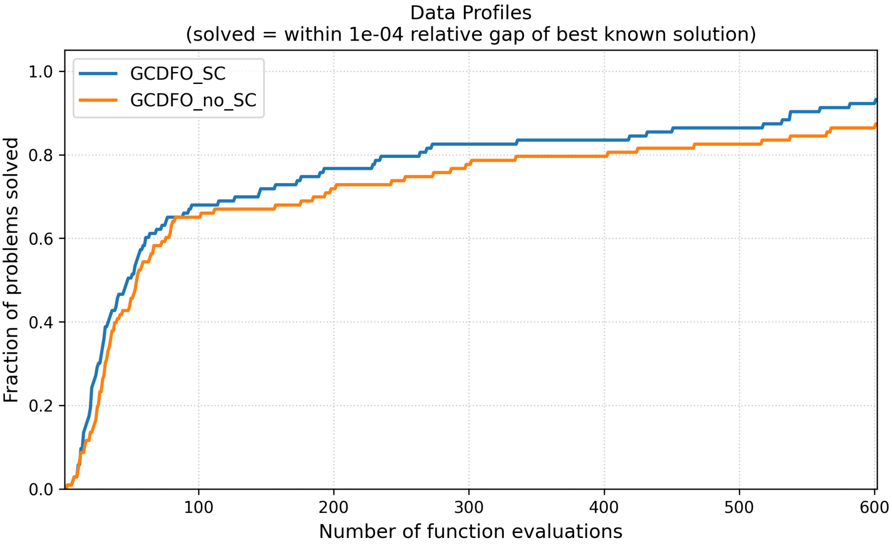
Complexity

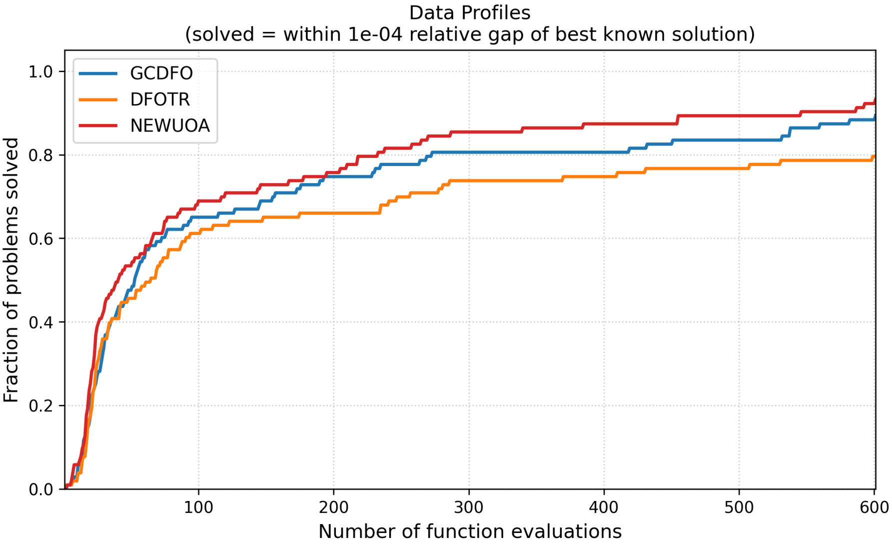
Theorem. Let $\mathcal{P}$ be the set of linear polynomials. Then the number of oracle calls in consecutive unsuccessful iterations is at most $O(n \log n)$.

Note: Comparatively, GC-TR has at most $3n$ consecutive unsuccessful iterations.

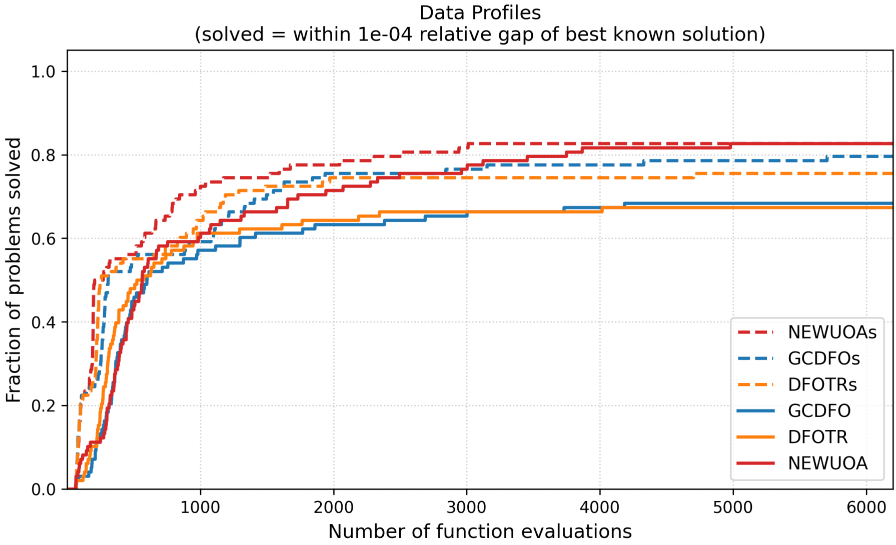
Is Self Correcting Worth It? (Low Dimensional)



Algorithm Comparison (Low Dimensional)



Algorithm Comparison (High Dimensional Subspace)



Conclusion

Summary

- Proposed a concrete geometry correcting trust region algorithm with flexible Lagrange polynomial basis choice
- Demonstrated empirical improvement when using geometry/self correcting steps
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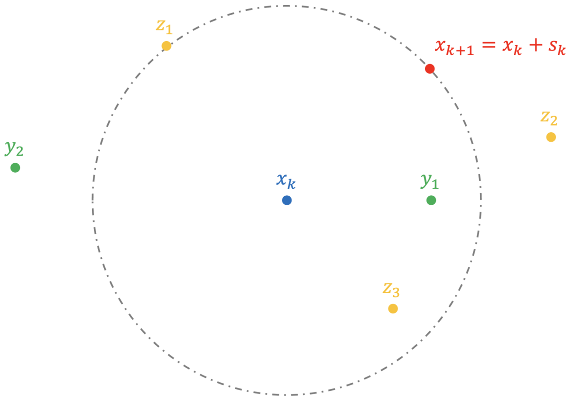
Future Directions

- Application of GCTR in randomized subspace
- Investigating different ways to fit a model using $\mathcal{Y}$ and $\mathcal{Z}$.

Thank you

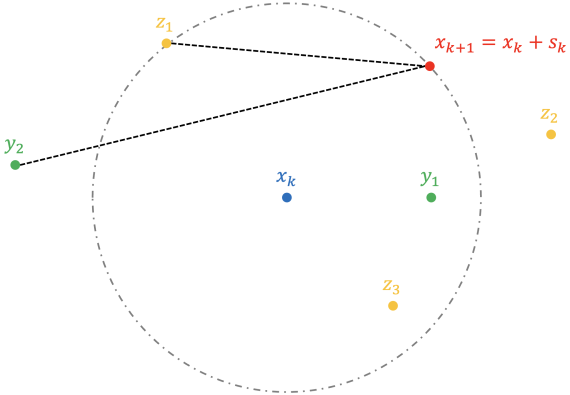
[CS25] Abraar Chaudhry and Katya Scheinberg. *On Complexity of Model-Based Derivative-Free Methods*. 2025. arXiv: 2510.14935 [math.OC]. URL: https://arxiv.org/abs/2510.14935.

Successful Subroutine



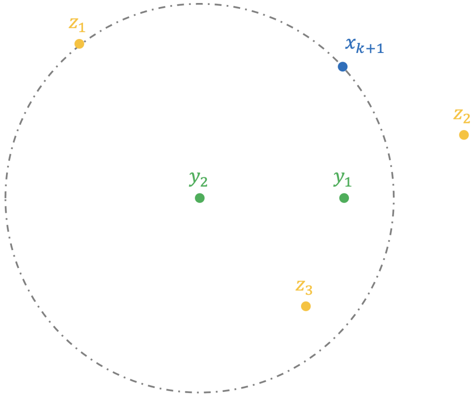
Accept new point and expand Δ_k.

Successful Subroutine



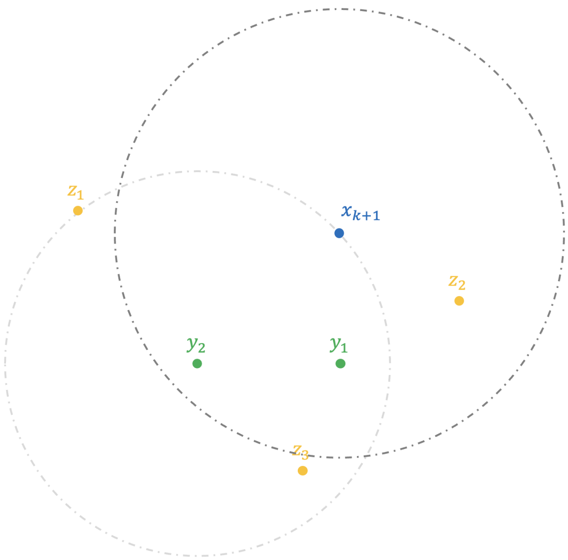
Compute max distance from new point to points in $\mathcal{Y}$ and $\mathcal{Z}$.

Successful Subroutine



Replace point in $\mathcal{Y}$ with old center.

Successful Subroutine



Ready for next iteration.

Algorithm Comparison (High Dimensional)